BIRZEIT UNIVERSITY

Faculty of Engineering & Technology Electrical & Computer Engineering Department

ENEE4113 - COMMUNICATION LAB
Experiment #3 PRELAB

Frequency Modulation

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Section: 4

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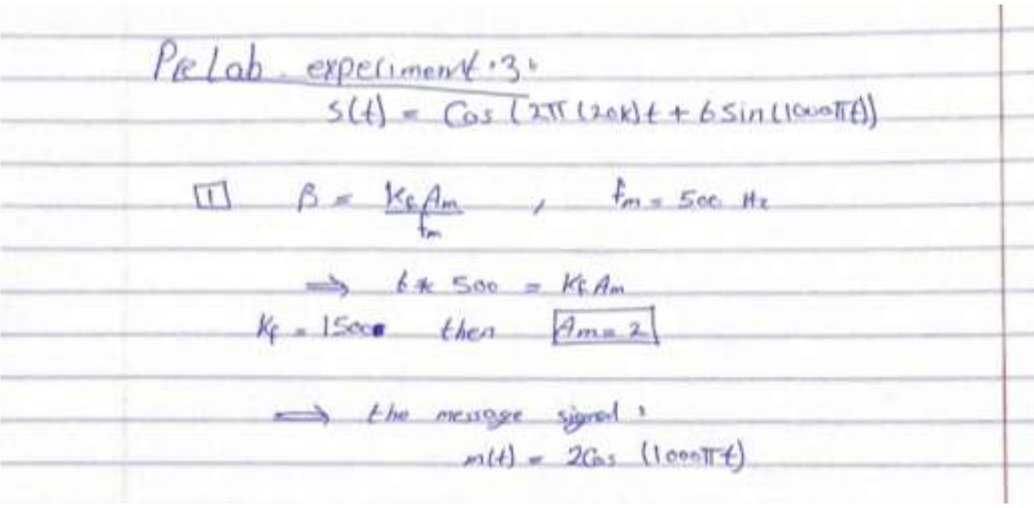
Extract the message signal m(t)



Prelab experiment 3:

$s(t) = \cos(2\pi(20k)t + 6\sin(1000\pi t))$

[1] $\beta = \frac{K_f A_m}{f_m}$, $f_m = 500$ Hz

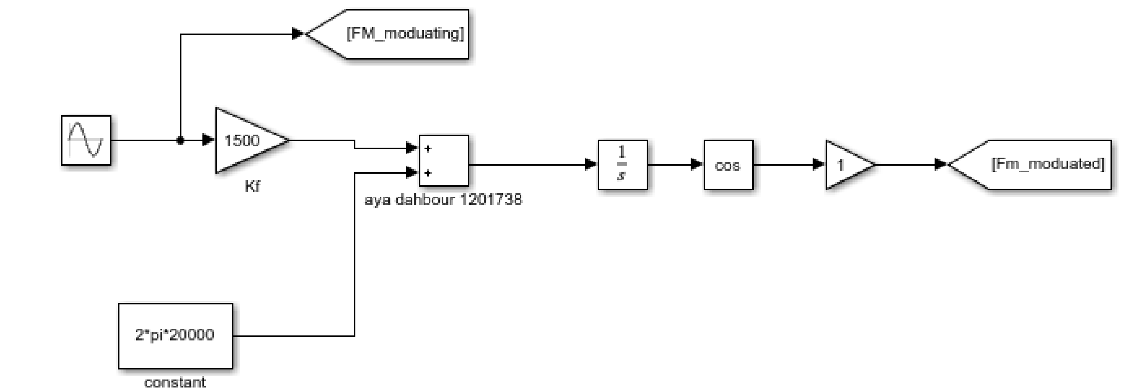
$\Longrightarrow 6 * 500 = K_f A_m$

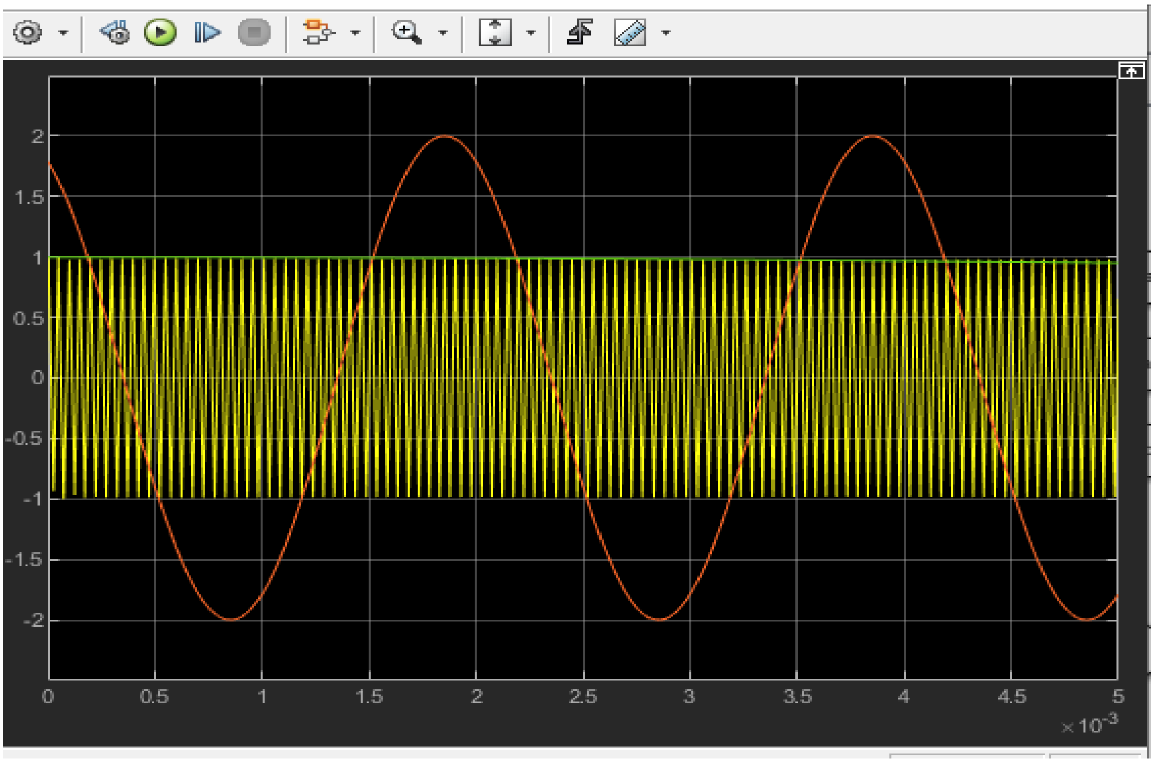
$K_f = 1500$ then $\boxed{A_m = 2}$

$\Longrightarrow$ the message signal:

$m(t) = 2\cos(1000\pi t)$

Plot message signal m(t) and s(t) versus t for -1 ≤ t ≤ 1





Extract message signal by using a phase-locked loop (PLL):